Quantum Computing

Noise and Quantum Error Correction

Lecture 6 Notes

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About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (6 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

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Chapter 1

Noise and Quantum Error Correction

Lecturer: Ralph Razzouk

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A real qubit is never isolated, and its state is disturbed by the environment. Measuring the qubit to find out what went wrong would destroy the very superposition we wish to protect. This lecture introduces the density matrix as the description of a qubit whose preparation is uncertain, models noise as bit flips and phase flips applied at random, and shows how the three-qubit code lets a parity measurement reveal the location of an error without revealing the encoded state. It closes with the limits of that code: which errors it corrects, which it cannot see, and what it silently assumes about the correction circuit itself.

1.1 Density matrices and noise

Every state in the previous five lectures was a vector: a preparation known exactly. In **??** we met a preparation that is not of this kind. A fair coin is flipped, and the qubit is prepared in $|0\rangle$ or $|1\rangle$ according to the result, which we are not told. No single vector describes this situation, since the qubit is in $|0\rangle$ half of the time and in $|1\rangle$ the other half. Noise produces exactly this kind of uncertainty: an error occurs or does not occur, with some probability, and we do not know which. We therefore need a description of a qubit whose preparation is uncertain, and we introduce it in the minimal form that this lecture requires.

1.1.1 Mixtures and coherences

Suppose the qubit is prepared in the state $|\psi_i\rangle$ with probability p_i, for some list of states and probabilities. Measured in a basis containing $|u\rangle$, the outcome u has probability $\sum_i p_i |\langle u|\psi_i\rangle|^2$, by the Born rule applied to each case and averaged over the cases. Writing $|\langle u|\psi_i\rangle|^2 = \langle u|\,|\psi_i\rangle\langle\psi_i|\,|u\rangle$, the sum becomes $\langle u|\,\rho\,|u\rangle$ for a single matrix ρ, which carries everything about the preparation that any measurement can access.

Definition 1.1: Density matrix

A qubit prepared in the state $|\psi_i\rangle$ with probability p_i, where $p_i \geqslant 0$ and $\sum_i p_i = 1$, is described by its density matrix

$$\rho = \sum_i p_i\, |\psi_i\rangle\langle\psi_i|\,. \tag{1.1}$$

If a single state occurs with probability one, $\rho = |\psi\rangle\langle\psi|$ and the preparation is called pure; otherwise it is called a mixture. Measuring in a basis containing $|u\rangle$ yields the outcome u with probability $\mathrm{tr}(\rho P_u)$, where $P_u = |u\rangle\langle u|$ is the projector onto $|u\rangle$.

The trace formula is the Born rule in a form that applies to pure states and mixtures alike, since $\mathrm{tr}(\rho P_u) = \langle u|\,\rho\,|u\rangle$ and, for a pure state, this is $|\langle u|\psi\rangle|^2$. The matrix ρ is Hermitian and has trace one, since $\mathrm{tr}\,|\psi_i\rangle\langle\psi_i| = \langle\psi_i|\psi_i\rangle = 1$. For a single qubit it is a 2×2 matrix

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}, \qquad \rho_{00} + \rho_{11} = 1, \qquad \rho_{10} = \rho_{01}^{*}. \tag{1.2}$$

The diagonal entries are the probabilities of the outcomes 0 and 1 in the computational basis, since $\mathrm{tr}(\rho P_0) = \rho_{00}$. They are called the populations. The off-diagonal entries are called the coherences, and in the computational basis they are what distinguishes a superposition of $|0\rangle$ and $|1\rangle$ from a mixture of $|0\rangle$ and $|1\rangle$.

Compare the superposition $|+\rangle$ with the coin-flip mixture of **??**. For the pure state, $|+\rangle\langle+|$ has every entry equal to $\frac{1}{2}$. For the mixture, the two outer products $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ are the two diagonal projectors, and their equal-weight average is half the identity:

$$\rho_{+} = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \rho_{\mathrm{coin}} = \frac{1}{2}\,|0\rangle\langle 0| + \frac{1}{2}\,|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{\mathbb{I}}{2}. \tag{1.3}$$

The two matrices agree on the diagonal, which is why the computational-basis measurement cannot tell them apart, and differ off the diagonal, which is why the $\{|+\rangle, |-\rangle\}$ measurement can. Evaluating $\mathrm{tr}(\rho P_+)$ gives 1 for ρ_+ and $\frac{1}{2}$ for ρ_{coin}, reproducing the argument of the box on superposition and ignorance in **??**; the calculation is written out below. The distinction that took a change of basis to exhibit in that lecture is now visible in a single matrix: a superposition of the computational basis states has coherences in that basis, and a mixture of those same states does not. The statement is basis-dependent; a mixture of $|+\rangle$ and $|-\rangle$ with unequal weights has nonzero coherences in the computational basis, and the pure state $|0\rangle$ has none. What distinguishes a pure state from a mixture in every basis is $\mathrm{tr}(\rho^2)$, which equals 1 only for pure states.

With $|+\rangle = \frac{1}{\sqrt{2}}(1,1)^{\mathsf{T}}$ and $|-\rangle = \frac{1}{\sqrt{2}}(1,-1)^{\mathsf{T}}$, the outer products are

$$\rho_{+} = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \rho_{-} = |-\rangle\langle-| = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \rho_{\mathrm{coin}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1.4}$$

All three have populations $\frac{1}{2}, \frac{1}{2}$, so $\mathrm{tr}(\rho P_0) = \rho_{00} = \frac{1}{2}$ for each, and the computational-basis measurement does not distinguish them. The coherences are $+\frac{1}{2}$, $-\frac{1}{2}$, and 0 respectively. The

projector onto $|+\rangle$ is $P_+ = |+\rangle\langle+| = \rho_+$, and

$$\operatorname{tr}(\rho_+ P_+) = \operatorname{tr}(P_+^2) = \operatorname{tr} P_+ = 1, \qquad \operatorname{tr}(\rho_- P_+) = \frac{1}{4}\operatorname{tr}\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0, \qquad \operatorname{tr}(\rho_{\text{coin}} P_+) = \frac{1}{2}\operatorname{tr} P_+ = \frac{1}{2}, \tag{1.5}$$

where $P_+^2 = P_+$ because P_+ is a projector, and the middle product vanishes because $\langle+|-\rangle = 0$. These are the three results of **??**: $|+\rangle$ gives + with certainty, $|-\rangle$ never does, and the coin flip gives + half of the time. In general $\operatorname{tr}(\rho P_+) = \langle+|\,\rho\,|+\rangle = \frac{1}{2}(\rho_{00} + \rho_{11} + \rho_{01} + \rho_{10}) = \frac{1}{2} + \operatorname{Re}\rho_{01}$, which shows directly that the X measurement reads the coherence and the computational-basis measurement reads the populations.

1.1.2 Dephasing

Definition 1.2: Dephasing

Dephasing is any process that reduces the coherences ρ_{01} and ρ_{10} while leaving the populations ρ_{00} and ρ_{11} unchanged. Complete dephasing sets the coherences to zero and turns $|+\rangle\langle+|$ into $\mathbb{I}/2$.

Dephasing is one of the principal things noise does to a superposition; the other, relaxation toward a preferred state such as $|0\rangle$, changes the populations as well and is not modelled in this course. The bit-flip model below is the simplest error that changes populations, although it drives them toward $\frac{1}{2}$ rather than toward a preferred state. A qubit whose relative phase is randomly disturbed by its surroundings, by a fluctuating field, or by an uncontrolled interaction, retains its populations but loses the definite relative phase that made it a superposition, and its density matrix drifts toward the diagonal. After complete dephasing, $|+\rangle$ has become indistinguishable from a coin flip. Every interference effect in **??** and **??** depends on relative phases, so dephasing is fatal to computation, and a quantum computer must either prevent it or undo it.

1.1.3 Error models

The simplest model of a discrete error uses the gates of **??**. A bit-flip error applies X to the qubit, exchanging $|0\rangle$ and $|1\rangle$. A phase-flip error applies Z, reversing the sign of $|1\rangle$ and hence the relative phase. A phase flip that occurs with probability p is dephasing in the sense just defined: the resulting density matrix is $(1-p)\rho + p\,Z\rho Z$, whose coherences are those of ρ multiplied by $1-2p$, as verified below; this is the one noise channel we write down explicitly, and the general theory of channels lies outside this course. The third Pauli matrix $Y = \mathrm{i}XZ$ is a bit flip and a phase flip together; it plays no role until section 1.3.

For the phase-flip model, $Z\rho Z$ has entries $(Z\rho Z)_{jk} = (-1)^j(-1)^k\rho_{jk}$, so the populations are unchanged and the coherences change sign. Therefore

$$(1-p)\rho + p\,Z\rho Z = \begin{pmatrix} \rho_{00} & (1-2p)\rho_{01} \\ (1-2p)\rho_{10} & \rho_{11} \end{pmatrix}, \tag{1.6}$$

which is dephasing in the sense of Definition 1.2. At $p = \frac{1}{2}$ the coherences vanish and ρ_+ becomes ρ_{coin}. Repeated application multiplies the coherences by $(1-2p)$ each time, so they decay geometrically.

Definition 1.3: Restricted bit-flip error model

Each qubit of a register is, independently of the others, either left alone with probability $1-p$ or subjected to a bit flip X with probability p. Errors occur once, after the state has been prepared and before it is examined.

This model is deliberately narrow. It contains one type of error, it treats qubits independently, and it places every error in a single time slot. It is enough to exhibit the mechanism of error correction, and section 1.3 explains what is left out and what removing each assumption costs.

1.2 The bit-flip code

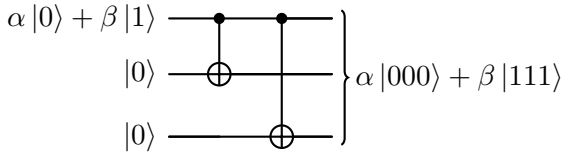
Classical error correction protects a bit by repetition: send 000 for 0 and 111 for 1, and a single flipped bit is outvoted by the other two. The quantum version cannot copy, by the no-cloning theorem of **??**, and cannot read the bits to take a vote without destroying a superposition. Both obstacles are avoided by the same construction.

1.2.1 Encoding

Definition 1.4: Code and logical qubit

A quantum code for one qubit is a two-dimensional subspace of the state space of n physical qubits, spanned by two orthonormal states $|0_L\rangle$ and $|1_L\rangle$, the logical basis states. A logical qubit is a state $\alpha|0_L\rangle + \beta|1_L\rangle$ in this subspace, called the code space. The three-qubit bit-flip code has $n = 3$, $|0_L\rangle = |000\rangle$, and $|1_L\rangle = |111\rangle$.

Encoding maps an unknown single-qubit state $\alpha|0\rangle + \beta|1\rangle$, together with two fresh qubits in $|0\rangle$, to the logical state $\alpha|000\rangle + \beta|111\rangle$. Two CNOT gates from **??** do this, both controlled by the first qubit and targeting the second and the third:



The term $\alpha|000\rangle$ is untouched by both gates, since the control is 0; the term $\beta|100\rangle$ acquires a 1 on the second qubit and then on the third. Linearity does the rest, and the verification is as follows.

Number the qubits $1, 2, 3$ from the top wire, and write CNOT_{12} for the gate controlled by qubit 1 acting on qubit 2. The circuit applies CNOT_{12} and then CNOT_{13}, which is the operator product $\mathrm{CNOT}_{13}\,\mathrm{CNOT}_{12}$. On the input $(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle = \alpha|000\rangle + \beta|100\rangle$,

$$\alpha|000\rangle + \beta|100\rangle \xrightarrow{\mathrm{CNOT}_{12}} \alpha|000\rangle + \beta|110\rangle \xrightarrow{\mathrm{CNOT}_{13}} \alpha|000\rangle + \beta|111\rangle\,. \tag{1.7}$$

Both gates act as the identity on $|000\rangle$ because the control qubit is 0, and each flips one target on the second term because the control qubit is 1. The map is linear in (α, β), as a unitary must be, and it sends the computational basis of the first qubit to the logical basis. For comparison, $(\alpha|0\rangle + \beta|1\rangle)^{\otimes 3}$ expands to eight terms with coefficients $\alpha^3, \alpha^2\beta, \ldots, \beta^3$, and equals the encoded state only when $\alpha\beta = 0$; this is the content of the box on cloning.

Wrong Concept 1.1: Error correction needs copies, which no-cloning forbids

The encoded state $\alpha|000\rangle + \beta|111\rangle$ is not three copies of $\alpha|0\rangle + \beta|1\rangle$. Three copies would be $(\alpha|0\rangle + \beta|1\rangle)^{\otimes 3}$, which contains terms such as $\alpha^2\beta|001\rangle$ that the encoded state lacks. The encoding is a unitary acting on the unknown qubit and two known ones; no-cloning forbids nothing of the kind. What it produces is an entangled state of three qubits in which no single qubit carries the relative phase of α and β on its own, as **??** showed for the Bell states; each qubit alone reveals only the populations $|\alpha|^2$ and $|\beta|^2$. Redundancy is spread across the

register rather than duplicated within it.

1.2.2 Syndrome measurement

Now suppose a bit flip strikes one qubit. Writing X_j for X applied to qubit j and the identity to the others, the possible states are

$$\alpha\,|000\rangle + \beta\,|111\rangle\,, \quad X_1\colon \alpha\,|100\rangle + \beta\,|011\rangle\,, \quad X_2\colon \alpha\,|010\rangle + \beta\,|101\rangle\,, \quad X_3\colon \alpha\,|001\rangle + \beta\,|110\rangle\,. \tag{1.8}$$

The four states live in four mutually orthogonal two-dimensional subspaces, one for each error, and the amplitudes α and β ride along unchanged in each. A measurement that identifies which of the four subspaces the state occupies, and nothing more, would locate the error without touching α and β. Such a measurement exists, and it asks about parities.

The measurements of **??** were made in an orthonormal basis, and we now use the same rule in a coarser form. A Hermitian operator whose eigenvalues are $+1$ and -1 splits the state space into two orthogonal eigenspaces, and to measure it is to perform a two-outcome measurement that distinguishes these eigenspaces: the outcome is the eigenvalue, its probability is the squared norm of the component of the state in that eigenspace, and the state is left projected onto that eigenspace and renormalized. This has the same outcome probabilities as the basis measurement of **??** with the basis vectors grouped into two sets, but not the same post-measurement state: it preserves superpositions within each eigenspace, which a full basis measurement would destroy, and that preservation is the whole point of what follows. Section 1.2.3 shows how such a measurement is carried out with the gates already known.

The operator Z_1Z_2, meaning Z on qubits 1 and 2 and the identity on qubit 3, has eigenvalue $+1$ on every computational basis state in which qubits 1 and 2 agree and -1 on every state in which they differ, because $Z\,|x\rangle = (-1)^x\,|x\rangle$. Measuring Z_1Z_2 therefore asks a single question, whether qubits 1 and 2 agree, and returns ± 1. It does not ask what their values are. Likewise Z_2Z_3 asks whether qubits 2 and 3 agree. In each of the four states in (1.8), both terms of the superposition give the same answer to both questions, so each state is an eigenvector of both operators and the two answers are certain.

In detail, since $Z\,|x\rangle = (-1)^x\,|x\rangle$ for $x \in \{0,1\}$, the operator $Z_1Z_2 = Z \otimes Z \otimes \mathbb{I}$ acts on a computational basis state as

$$Z_1Z_2\,|x_1x_2x_3\rangle = (-1)^{x_1+x_2}\,|x_1x_2x_3\rangle = (-1)^{x_1\oplus x_2}\,|x_1x_2x_3\rangle\,, \tag{1.9}$$

with eigenvalue $+1$ when $x_1 = x_2$ and -1 otherwise. On $|000\rangle$ and $|111\rangle$ the eigenvalue is $+1$; on $|100\rangle$, $|011\rangle$, $|010\rangle$, $|101\rangle$, where exactly one of the first two qubits has been flipped relative to a code state, it is -1; on $|001\rangle$ and $|110\rangle$ it is $+1$ again. The same reasoning with $Z_2Z_3 = \mathbb{I} \otimes Z \otimes Z$ produces the two syndrome columns of Table 1.1.

Definition 1.5: Syndrome

The syndrome of a state is the pair of outcomes $(Z_1Z_2, Z_2Z_3) \in \{+1,-1\}^2$ obtained by measuring the two parity checks. The code space is the subspace on which both checks give $+1$.

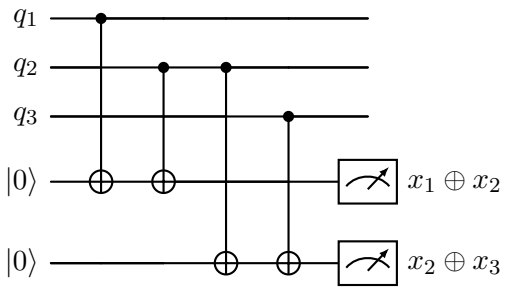
Wrong Concept 1.2: Measuring the syndrome collapses the superposition

The projection rule of **??** applies to the observable being measured. The state $\alpha|100\rangle+\beta|011\rangle$ is an eigenvector of Z_1Z_2 with eigenvalue -1: both terms have qubits 1 and 2 disagreeing. The measurement therefore returns -1 with certainty and, since the state is already an eigenvector, leaves it exactly as it was. Nothing is learned about α or β, because the outcome would have been the same for every value of them. A measurement collapses only what it distinguishes. The parity checks are designed to distinguish the four error subspaces from one another and to distinguish nothing within any of them.

1.2.3 Measurement circuit

In a circuit, each parity check is performed by two CNOT gates from the two data qubits into a fresh ancilla qubit prepared in $|0\rangle$, followed by a computational-basis measurement of the ancilla. The ancilla ends in $|x_1 \oplus x_2\rangle$, whose measurement gives 0 for eigenvalue $+1$ and 1 for eigenvalue -1, and the data qubits are left untouched. This is the central idea of the lecture in operational form: the syndrome is read from the ancillas, and the logical information remains in the data qubits, unmeasured.

The following circuit measures both checks. Two ancillas, drawn as the lower wires, start in $|0\rangle$.



On a computational basis state $|x_1x_2x_3\rangle|0\rangle|0\rangle$ the first two CNOT gates add x_1 and then x_2 modulo 2 into the first ancilla, and the last two add x_2 and x_3 into the second, giving $|x_1x_2x_3\rangle|x_1 \oplus x_2\rangle|x_2 \oplus x_3\rangle$. Now take a state $\alpha|x\rangle+\beta|y\rangle$ from Table 1.1, in which the strings x and y have the same parities $s_1 = x_1 \oplus x_2 = y_1 \oplus y_2$ and $s_2 = x_2 \oplus x_3 = y_2 \oplus y_3$. By linearity the circuit produces

$$\big(\alpha|x\rangle+\beta|y\rangle\big)|s_1\rangle|s_2\rangle, \tag{1.10}$$

a product state in which the ancillas carry the syndrome and the data qubits carry the original superposition. Measuring the ancillas gives s_1s_2 with certainty and, because the state is a product, leaves the data qubits exactly as they were. The ancilla value 0 corresponds to the eigenvalue $+1$ and the value 1 to -1. This is the operational form of the statement that a syndrome measurement acquires no information about α and β.

1.3 Correction and its limits

1.3.1 Recovery

The four states in (1.8) have four distinct syndromes, listed in Table 1.1. Since $X_jX_j = \mathbb{I}$, the error is undone by applying X to the qubit that the syndrome names.

Error	State after error	Z_1Z_2	Z_2Z_3	Recovery
none	$\alpha\|000\rangle + \beta\|111\rangle$	$+1$	$+1$	do nothing
X_1	$\alpha\|100\rangle + \beta\|011\rangle$	-1	$+1$	apply X_1
X_2	$\alpha\|010\rangle + \beta\|101\rangle$	-1	-1	apply X_2
X_3	$\alpha\|001\rangle + \beta\|110\rangle$	$+1$	-1	apply X_3

Table 1.1: The syndrome table of the three-qubit bit-flip code. The syndrome depends on the error and not on the amplitudes α, β.

Example 1.1 (Reading one syndrome)

The two parity checks are measured and return $Z_1Z_2 = +1$ and $Z_2Z_3 = -1$. Qubits 1 and 2 agree, and qubits 2 and 3 disagree, so qubit 3 is the odd one out. Table 1.1 gives the error X_3 and the recovery X_3, which returns the state to $\alpha|000\rangle + \beta|111\rangle$. The amplitudes were never measured and are not known, before or after. As a second observation, a phase flip Z_2 turns the encoded state into $\alpha|000\rangle - \beta|111\rangle$, on which both checks return $+1$. The table reports no error, and none is corrected.

The first part of the example shows that a syndrome identifies a location; the second shows that the same table is blind to a phase flip, which is the limitation taken up in section 1.3.3.

Proposition 1.1 Correction of a single bit flip

Let $\alpha|000\rangle + \beta|111\rangle$ be subjected to at most one bit flip, on an unknown qubit. Measuring Z_1Z_2 and Z_2Z_3 and applying the recovery in Table 1.1 returns the state to $\alpha|000\rangle + \beta|111\rangle$ with certainty, for every α and β.

Proof: Each row of Table 1.1 lists a state whose two terms are computational basis states with the same pair of parities, so the state is an eigenvector of both checks and the measurement returns the listed syndrome with certainty and leaves the state unchanged. The four syndromes are distinct, so the syndrome determines the row, and the listed recovery is the inverse of the listed error, since $X_j^2 = \mathbb{I}$. Nothing in the argument depends on α or β. ■

1.3.2 Failure probability

The proposition says nothing about two or more flips, and there the code fails. After X_1X_2 the state is $\alpha|110\rangle + \beta|001\rangle$, with syndrome $(+1, -1)$, the same as for X_3. The recovery applies X_3 and produces $\alpha|111\rangle + \beta|000\rangle$: a valid code state, with the logical basis states exchanged. This is a logical error, and it is worse than an uncorrected physical error because the code reports success. Three flips give the same result with syndrome $(+1, +1)$. Under Definition 1.3 the probability of two or more flips among three qubits is

$$p_{\text{fail}} = 3p^2(1-p) + p^3 = 3p^2 - 2p^3, \tag{1.11}$$

which is below p exactly when $p < \frac{1}{2}$, as verified below. For $p = 10^{-2}$ the encoded qubit fails with probability about 3×10^{-4}. The code trades one qubit for three and replaces a failure probability of order p by one of order p^2; larger codes replace it by higher powers. The threshold $p < \frac{1}{2}$ is a property of this code under this error model, not a general statement about quantum hardware.

Under Definition 1.3 the number of flipped qubits is binomial with three trials and success probability p. The code corrects zero or one flip and fails for two or three, so

$$p_{\text{fail}} = \binom{3}{2} p^2(1-p) + \binom{3}{3} p^3 = 3p^2 - 3p^3 + p^3 = 3p^2 - 2p^3. \tag{1.12}$$

The inequality $p_{\text{fail}} < p$ reads $3p^2 - 2p^3 < p$, or, dividing by $p > 0$, $2p^2 - 3p + 1 > 0$, that is $(2p-1)(p-1) > 0$. For $0 < p < 1$ the second factor is negative, so the product is positive exactly when $2p - 1 < 0$, that is $p < \frac{1}{2}$. At $p = \frac{1}{2}$ both sides equal $\frac{1}{2}$, and above it encoding makes matters worse. For small p, $p_{\text{fail}} \approx 3p^2$: with $p = 10^{-2}$, $p_{\text{fail}} = 2.98 \times 10^{-4}$, and with $p = 10^{-3}$, $p_{\text{fail}} \approx 3 \times 10^{-6}$.

The two-flip failure is a logical error rather than a detected failure. After X_1X_2 the state is $\alpha\,|110\rangle + \beta\,|001\rangle$; the parities are $1 \oplus 1 = 0$ and $1 \oplus 0 = 1$, so the syndrome is $(+1, -1)$, the recovery is X_3, and the result is $\alpha\,|111\rangle + \beta\,|000\rangle = \alpha\,|1_L\rangle + \beta\,|0_L\rangle$. The logical qubit has suffered a logical X, and no record of the event exists.

Remark 1.1 Digitization of noise

The syndrome measurement has four possible outcomes, and each is followed by one of four recoveries. There is no estimation of a continuous parameter and no gradual accumulation of small deviations within a run of the procedure: the state is projected onto one of the four error subspaces, and for any error in the span of $\mathbb{I}$, X_1, X_2, X_3, including a small continuous rotation of one qubit, the recovery matched to whichever subspace results is exact. Failure comes only from error components outside that span, such as two simultaneous flips. This is the sense in which quantum error correction digitizes noise. It is also why the assumptions of the model, that the measurement and the recovery are themselves perfect, carry so much weight.

1.3.3 Phase errors

The second illustration of Example 1.1 exposes the code's blind spot. A phase flip Z_j on any single qubit maps $\alpha\,|000\rangle + \beta\,|111\rangle$ to $\alpha\,|000\rangle - \beta\,|111\rangle$, because $Z\,|0\rangle = |0\rangle$ and $Z\,|1\rangle = -\,|1\rangle$ and the second term has a 1 in every position. The result lies in the code space with both checks at $+1$, so it is undetected, and it is a logical phase flip, since the relative sign between $|0_L\rangle$ and $|1_L\rangle$ has been reversed. The code offers no protection against Z errors, and since any of three qubits can suffer one, an encoded qubit is roughly three times as exposed to phase flips as an unencoded one.

In detail, take $Z_2 = \mathbb{I} \otimes Z \otimes \mathbb{I}$. Since $Z\,|0\rangle = |0\rangle$ and $Z\,|1\rangle = -\,|1\rangle$,

$$Z_2\big(\alpha\,|000\rangle + \beta\,|111\rangle\big) = \alpha\,|000\rangle - \beta\,|111\rangle\,, \tag{1.13}$$

and the same holds for Z_1 and Z_3, since each code basis state has all three qubits equal. The result is a code state, so both parity checks return $+1$ and Table 1.1 prescribes no action. Relative to the logical basis, the state has changed from $\alpha\,|0_L\rangle + \beta\,|1_L\rangle$ to $\alpha\,|0_L\rangle - \beta\,|1_L\rangle$, which is a logical phase flip Z_L. Applying Z to any odd number of the three qubits has the same effect, and applying it to an even number has none, so the code space is preserved by every product of Z operators, which is why no parity check built from Z operators can see them.

The remedy follows from the identity $HZH = X$ of **??**: in the $\{|+\rangle\,, |-\rangle\}$ basis a phase flip is a bit flip. Encoding into $|0_L\rangle = |{+}{+}{+}\rangle$ and $|1_L\rangle = |{-}{-}{-}\rangle$, obtained by applying H to each qubit of the bit-flip code, and replacing the checks by X_1X_2 and X_2X_3, gives a three-qubit phase-flip code that corrects any single Z and is blind to X. Neither code alone suffices, since real noise contains both, and combining the two constructions is the route to codes that correct both.

In the phase-flip code the roles are exchanged: with $|0_L\rangle = |{+}{+}{+}\rangle$ and $|1_L\rangle = |{-}{-}{-}\rangle$, a Z_j maps $|+\rangle$ to $|-\rangle$ on qubit j and is detected by the checks X_1X_2 and X_2X_3, since $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$, while an X_j leaves both logical states unchanged up to a sign and passes undetected.

Remark 1.2 General codes

Every 2×2 matrix is a linear combination of $\mathbb{I}$, X, Y, and Z. A code that detects and corrects each of X_j, Y_j, Z_j for every single qubit j, by the same mechanism of distinct syndromes on orthogonal error subspaces, corrects every possible single-qubit error, because the syndrome measurement projects any superposition of these four cases onto one of them. This is the design goal of the general theory, and the codes that meet it, such as Shor's nine-qubit code [1] and the seven-qubit Steane code [2], both members of the CSS and stabilizer families, are named here as directions for further study rather than material of this course. What they share with the three-qubit code is the structure already visible here: a set of commuting checks, all equal to $+1$ on the code space, whose sign patterns label the correctable errors.

1.3.4 Commuting checks

The checks Z_1Z_2 and Z_2Z_3 commute with each other, since all Z operators on distinct or identical qubits commute, so they can be measured simultaneously and in either order with the same result. Each of them fixes both $|000\rangle$ and $|111\rangle$, and hence every state in the code space; the code space is precisely their joint $+1$ eigenspace, and two independent checks on three qubits cut the eight-dimensional space down to the two dimensions of one logical qubit. An error X_j anticommutes with every check that contains Z_j and commutes with every check that does not, because $XZ = -ZX$ on the same qubit. If S is a check and $SX_j = -X_jS$, then for any code state $|\psi_L\rangle$ we have $SX_j|\psi_L\rangle = -X_jS|\psi_L\rangle = -X_j|\psi_L\rangle$, so the corrupted state is an eigenvector of S with eigenvalue -1. The syndrome is therefore the pattern of commutation and anticommutation between the error and the checks, which explains at once why it is independent of α and β and why Z_j, which commutes with every check, produces the trivial syndrome. A set of commuting operators that fix the code space is called a stabilizer, and codes designed by choosing such a set are stabilizer codes. We have named the idea; developing it is beyond the scope of this course.

1.3.5 Assumptions

Finally, the assumptions under which Proposition 1.1 holds must be stated plainly. The encoding circuit was perfect. The syndrome measurement, including the ancillas and the CNOT gates that write parities into them, was perfect. The recovery gate was perfect. Errors struck only the data qubits, only once, only between encoding and checking, and only as bit flips. In real hardware every one of these operations is itself noisy, a faulty CNOT can copy an error from one qubit to another, and a wrong syndrome can trigger a recovery that creates the very error it was meant to remove. Whether correction can still help, and under what conditions on the physical error rate, is the subject of **??**.

1.4 Summary

A qubit with an uncertain preparation is described by a density matrix, whose diagonal entries are the outcome probabilities in the computational basis and whose off-diagonal entries, the coherences, are the content of a superposition that a mixture lacks. Dephasing is the decay of the coherences, and it is one of the principal effects of noise on superpositions. Bit-flip and phase-flip errors are the gates X and Z applied at random. The three-qubit bit-flip code spreads one logical qubit over three physical qubits, without copying it, so that a single bit flip moves the state into one

of three orthogonal subspaces identified by the parity checks Z_1Z_2 and Z_2Z_3. The syndrome reveals the error location and nothing about the logical amplitudes, because the checks return the same value on both terms of the superposition. Recovery is exact for one flip and fails for two or more, with probability of order p^2. The code is blind to phase flips, which the Hadamard basis converts into bit flips, and a general code must correct any single-qubit Pauli error through commuting checks. All of this assumed perfect encoding, checks, and recovery.

Homework 3, part B, can be attempted after this lecture.

References

[1] Peter W. Shor. Scheme for reducing decoherence in quantum computer memory. *Physical Review A*, 52:R2493–R2496, 1995.

[2] Andrew M. Steane. Error correcting codes in quantum theory. *Physical Review Letters*, 77: 793–797, 1996.